

Exp No: 18

GENERATION OF PWM USING TIMER INTERRUPT

Objective :

1. To develop an Embedded C program for generating a Pulse Width Modulation (PWM) signal using a timer interrupt in the 8051 microcontroller.
2. To control the duty cycle of the PWM signal using timer interrupts.

Code:

```
#include <reg51.h>
```

```
sbit PWM = P1^0;
```

```
unsigned char count = 0;
```

```
unsigned char duty = 100; // Change this value (0–200)
```

```
void timer0_ISR(void) interrupt 1
```

```
{
```

```
TH0 = 0xFC; // Reload for 1ms delay
```

```
TLO = 0x66;
```

```
count++;
```

```
if(count < duty)
```

```
PWM = 1; // ON time
```

```
else
```

```
PWM = 0; // OFF time
```

```
if(count >= 200) // Total period = 200ms
```

```
count = 0;
```

```
}
```

```
void main()
```

```
{
```

```
TMOD = 0x01; // Timer0 Mode 1 (16-bit)
```

```
TH0 = 0xFC; // 1ms preload
```

```
TL0 = 0x66;
```

```
IE = 0x82; // EA = 1, ET0 = 1
```

```
TR0 = 1; // Start Timer0
```

```
while(1);
```

```
}
```

Output:

